# Combined Thermal Energy Storage and Dedicated Mechanical Subcooling for $CO_2$ refrigeration systems

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**Abstract**

Carbon dioxyde ($CO_2$) is increasingly adopted in commercial refrigeration systems thanks to its low environmental impact. However, its supercritical properties lead to important reduction in efficiency in transcritical cycles. Thus, this study investigates the combined integration of Dedicated Mechanical Subcooling (DMS) and Thermal Energy Storage (TES) to enhance energy and economic performance of these systems. A comprehensive method is developed in order to simultaneously determine the optimal sizing and operation strategy of a three-component system (main $CO_2$ cycle, DMS auxiliary cycle, and TES system) over a representative year. The approach couples thermodynamic modeling and pre-computation with Mixed-Integer Linear Programming (MILP). Applied to a supermarket refrigeration case in Madrid, the results show that implementing DMS alone achieves a 43.5% reduction in total cost and 15.4% energy savings by halving the main compressor capacity. Adding TES provides additional benefits, reaching 46.8% total cost reduction and 18.2% energy savings with regard to the reference case. Notably, the optimal share of DMS maximizing the overall instantaneous performance has to be capped so as to achieve a significant TES utilization and minimum energy consumption. The analysis also reveals that lower TES storage temperatures reduce total cost, despite decreasing charging efficiency, since maximizing availability of subcooling from TES during peak demand periodsthe is more valuable.
**Keywords:** $CO_2$ refrigeration, Dedicated Mechanical Subcooling, Thermal Energy Storage, system optimization, MILP

## 1. Introduction

In the era of global warming, the need for refrigeration consistently increases. Concurrently technologies evolve driven by technical progress and under the constraint of environmental regulations.

From the 1930s onward, for more than half a century, fluorinated gases and other halogenoalkanes were the standard refrigerants. Their advantages include relative harmlessness to humans and good thermodynamic properties. However, their detrimental impact on the ozone layer led to international agreements regulating, limiting, or even banning their use [1], [2]. The EU followed suit by putting in place a legal framework for refrigerants, and even by regulating more proactively [3].

One of the interesting alternatives is the $CO_2$ (R744). This natural and widely available refrigerant has no impact on the ozone layer and a Global Warming Potential (GWP) of 1. Being non-flammable, non-toxic and non-corrosive, it also an advantageous volumetric cooling capacity 3 to 10 times greater than that of common fluorinated gases, which makes it a promising solution for compact systems [4]. These strengths make this technology particularly attractive for applications in supermarkets, where operational safety and low-temperature performance (for frozen food displays and negative cold rooms) are required [5].

Apart from the quest for more environmentally friendly fluids, it is the technological advances allowing the confinement of gases under high pressure that have made it possible to reconsider R744 for multiple applications Actually, the critical point of $CO_2$ being (31.06 °C, 73.825 bar), many operating conditions result in a transcritical cycle and very high operating pressures (up to exceeding 100 bars).

Yet, in transcritical operations, despite the high heat rejection pressure, the lack of isothermal condensation reduces the efficiency of the heat transfer to the hot source. This results in the reduction of the Coefficient Of Performance (COP) of the cycle. Subcooling the fluid downstream of the gas-cooler then appears as a mean for improving the efficiency. By reducing the enthalpy of the fluid before expansion, it effectively increases the specific cooling capacity obtained during evaporation. Of course, this benefit provided by subcooling also applies to subcritical operation. Several methods can provide fluid subcooling, they differ in the level of energy gain as well as the complexity of the required system [6].

One of these methods is Dedicated Mechanical Subcooling (DMS) [7], [8]. It is based on a small mechanical vapour-compression cycle, which extracts heat from the $CO_2$ at the exit of the condenser/gas-cooler of the main refrigeration cycle. This promising solution have proved to enable increments in COP ranging from 7.8 % to 17.5 %, depending on the ambient air temperature [9].

Another method that can be considered relies on a Thermal Energy Storage (TES) system, which allows the demand for refrigeration to be shifted over time. With a wide range of possible technologies, its coupling with $CO_2$ cycles is considered a real lever for improving energy efficiency, particularly for commercial refrigeration systems [5]. This technology

would indeed make it possible to achieve up to 5.6 % in energy consumption reduction [10]. In this work, taking advantage of daily variations in ambient temperature is considered so as to produce and store cooling energy when production conditions are optimal. Then it is possible to deliver this energy stock at the most critical times of the day [11], [12], [13].

For commercial refrigeration applications, these $CO_2$ system performance enhancement devices have already been studied independently but never associated. The objective of this work is therefore to investigate the potential benefits of combining both DMS and TES to a $CO_2$ refrigeration cycle, with the particularity that the DMS system also charges the TES system, as its COP is better than that of the main cycle at $CO_2$.

Thus, the structure under scrutiny includes 3 components (see Figure 1), the design and operation of which are intended to be optimized. This work is based on a concrete case of study, which provides the input time series of external constraints (ambient air temperature, refrigeration load or electricity price signal) needed as support for optimization. A supermarket in Madrid (Spain), whose refrigeration load has been modeled by [14], has been chosen. The system must cover the refrigeration needs of the medium temperature (-10°C) refrigerated display cases.

**Method** (illustrated in Figure 2):

The goal is to buid and solve an optimization problem to minimize the total cost over of the system over several years. This total cost results from both investment and system operation. Thus, the objectives include both the proper sizing of the components (capacity of compressors, volume of storage tanks, etc.) and their optimal operation over time (power levels, operating modes, etc.). Given that the electricity price signal is considered, the energy consumed by the system will be minimized, relative to the variable availability of the resource.

The electrical power consumptions and heat flows are linked by the COPs of the different cycles, which must therefore be determined independently of their load, for the ambient air temperatures of the time series. This requires, first and foremost, a detailed model of the different thermodynamic cycles, of their coupling and of the allowed operating modes ( Section 2). However, there are degrees of freedom within the system, such as the heat rejection pressure in a transcritical cycle, which must be prior fixed to get the maximal performance ( Section 3). It is also necessary to ensure compliance with certain technical constraints in the computed cycles, especially the minimum pinch in heat exchangers ( Section 3.1).

In addition to the COPs, the optimization inputs are certain physical boundaries that thermodynamics imposes (Section 3.6), external constraints (like the refrigeration load, Section 4.1), as well the size-related costs of the components and the costs of electricity ( Section 4.3).

Prior to the formulation of the core optimization problem, the calculation of these parameters must be done. Some are time-independent, while others are time-dependent and require to be computed at each time step in order to get a time series.

Then, using these inputs, the core problem can then be formulated within the framework of linear optimization, and the solver can run on the time series ( Section 4).

## 2. Overall system model

In this work computations and simulations have been implemented with object-oriented programming in Python. The library `CoolProp` is used to get the thermodynamic properties of the fluids and the library `scipy` to solve non-linear problems or to perform binary searches. These tools enable to model the three-component system which has 5 different operating modes, the characteristics of which are detailed in Table 3.

### 2.1. Component characteristics

The thermodynamic cycle of the working fluid ($CO_2$, also referred to as R744) of the main component can be described with the stages of Table 1. It is worth noting that the subcooling stages ($3 \rightarrow 4$ and $4 \rightarrow 5$) are optional, there may be none, only one or both of them. It is also assumed that condensation cannot occur in a subcooler. This implies that in a subcritical cycle, the working fluid enters the first subcooler at the bubble point and that in a transcritical cycle, the working fluid both enters and exits the subcoolers in a supercritical state. Its thermodynamic cycle is represented in a P-h diagram in Figure 1.

| Step | Process | Contact with |
|---|---|---|
| $1 \rightarrow 2$ | Compression | - |
| $2 \rightarrow 3$ | Isobaric cooling/condensation | Ambient air |
| $3 \rightarrow 4$ | Possible isobaric subcooling | DMS system R290 |
| $4 \rightarrow 5$ | Possible isobaric subcooling | TES system water |
| $5 \rightarrow 6$ | Isenthalpic expansion | - |
| $6 \rightarrow 1$ | Isobaric evaporation | Refrigerated load |

**Table 1:** Main $CO_2$ cycle stages

The propane (R290) used in the DMS system has a subcritical refrigeration cycle between the ambient air and a heat source imposing $T_{evap\ prop}$. This latter can be the subcooled $CO_2$ or the water of the charging TES.

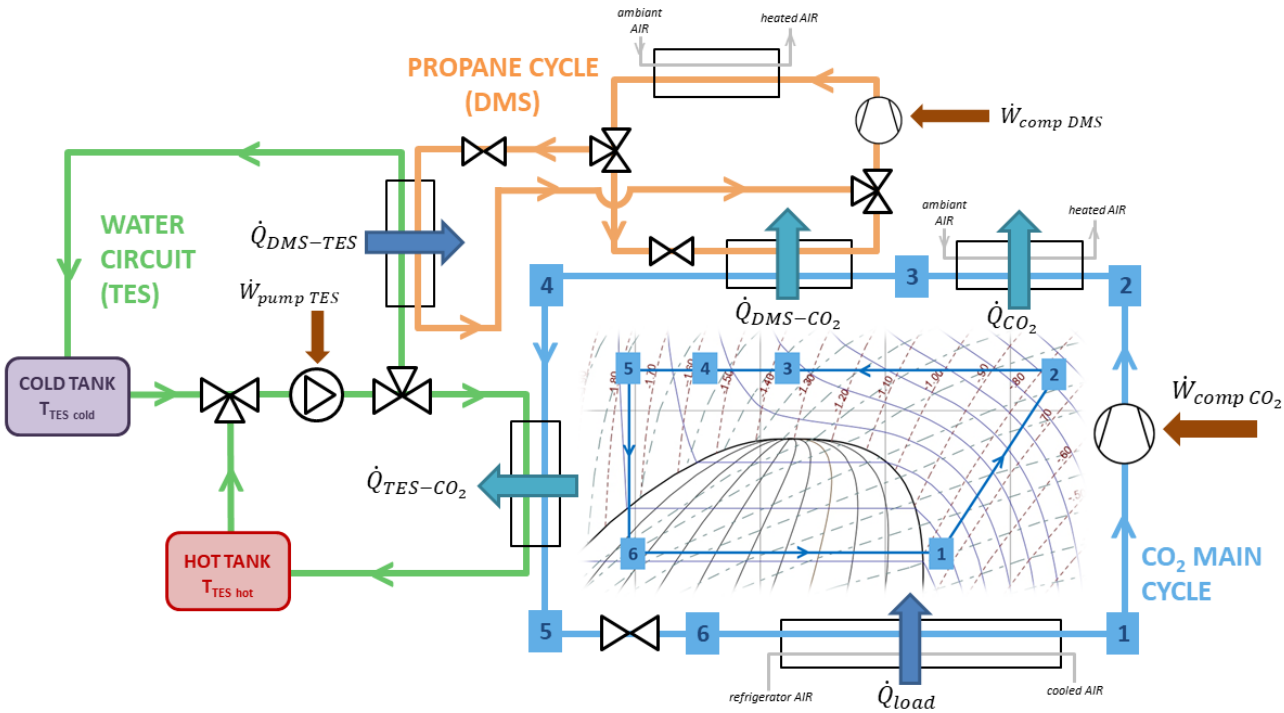
Once charged, the TES system can also provide subcooling to the main cycle ($4 \rightarrow 5$). This system uses sensible thermal energy storage, based on 2 perfectly insulated water tanks, whose temperature $T_{TES\ hot}$ and $T_{TES\ cold}$ are assumed uniform and constant. Heat exchangers are placed on the charging and discharging sides between the 2 tanks. Compared to latent or thermochemical TES, the technology used is the most economical, although less energy dense [12].

### 2.2. Couplings

The degrees of subcooling $\Delta T_{sc\ DMS}$ and $\Delta T_{sc\ TES}$ (expressed in $^\circ C$) provided to the $CO_2$ are cohensive coupling parameters between the main cycle and the other components. However, in order to ease the formulation of a linear optimization problem, energy-based coupling variables are preferred. Thus, the heat flow rates $\dot{Q}_{sc\ DMS}$ and $\dot{Q}_{sc\ TES}$ exchanged in the subcoolers are rather considered to define the two following ratios:

$$\alpha = \frac{\dot{Q}_{sc\ DMS}}{\dot{Q}_{CO_2}} = \frac{\Delta h_{sc\ DMS}}{\Delta h_{CO_2}} = \frac{h_3 - h_4}{h_1 - h_3}$$

**Figure 1:** Overall system architecture




| General | | |
|---|---|---|
| $\Delta T_{\text{pinch air}}$ | $5°C$ | |
| $\Delta T_{\text{pinch water}}$ | $3°C$ | |
| **Main $CO_2$ Cycle** | | |
| $T_{\text{evap } CO_2}$ | $-10°C$ | |
| $\Delta T_{\text{superheating } CO_2}$ | $5°C$ | |
| $\eta_{\text{comp } CO_2}(r)$ | $1.003 - 0.121 \cdot r$ | (from [15]) |
| **DMS System (using R290 = propane)** | | |
| $\Delta T_{\text{superheating prop}}$ | $7°C$ | |
| $\Delta T_{\text{subcooling prop}}$ | $1°C$ | |
| $\eta_{\text{comp prop}}(r)$ | $0.3774 + 0.1405 \cdot r - 0.0201 \cdot r^2 + 0.0008 \cdot r^3$ | (from [16]) |

**Table 2:** System fixed characteristics

$$\beta = \frac{\dot{Q}_{sc\ TES}}{\dot{Q}_{CO_2}} = \frac{\Delta h_{sc\ TES}}{\Delta h_{CO_2}} = \frac{h_4 - h_5}{h_1 - h_3}$$

### 2.3. *Performance measurement*

In order to distribute the refrigeration load properly during the optimization process, it is necessary to define performance indicators for the different thermodynamic cycles. It is done using the specific properties of the fluids, as follows:

$$\text{COP}_{CO_2} = \frac{h_1 - h_3}{h_2 - h_1} \quad \text{and} \quad \text{COP}_{\text{DMS}} = \frac{\Delta h_{ev\ prop}}{w_{comp\ prop}}$$

Moreover, the instantaneous overall performance of the refrigeration system is measured by the ratio between the total refrigeration power delivered and the total electrical power consumed by compressors. This coefficient can be written with the previous ones and the coupling parameters:

$$\text{COP}^* = \frac{\dot{Q}_{ev}}{\dot{W}_{total}} = \frac{\dot{Q}_{ev}}{\dot{W}_{comp\ CO_2} + \dot{W}_{comp\ DMS}}$$

$$\text{COP}^* = \frac{1 + \alpha + \beta}{1 + \alpha \cdot \frac{\text{COP}_{CO_2}}{\text{COP}_{\text{DMS}}}} \cdot \text{COP}_{CO_2}$$

## 3. Optimal thermodynamic settings

The internal degrees of freedom of the system must be fixed so that they maximize the overall instantaneous performance COP*. For this, tools from the `scipy` library are used to perform non-linear optimization (`optimize.minimize`) or to perform dichotomous searches (`optimize.bisect`).

Moreover, an important contribution of this work is to rigorously examine compliance with the minimum pinch requirements ($\Delta T_{pinch}$) in each of the heat exchangers ( Section 3.1), including during the optimization of other thermodynamic parameters of the cycles.

### 3.1. *Pinch requirements in heat exchangers*

In each heat exchanger, locating the pinch point and measuring the associated minimum temperature approach is essen-

tial to ensure that the pinch requirements are met. The feasibility of the designed cycles is at stake.

This motivates a robust implementation of the pinch constraints in the thermodynamic models. More information on this aspect are detailed in **Supplementary material - Pinch study**. For most of the heat exchangers of the system, the temperature profiles of the main working fluid has showed that the pinch point was located at the outlet or one end of the exchanger, which facilitates compliance with the pinch requirements. For instance, these are achieved with a suitable propane condensation temperature $T_{cond\ prop}$ or with an upper boundary imposed on $\beta$ to maintain a sufficient temperature difference between the $CO_2$ and the water of the TES. However, for the two exchangers using ambient air to cool a fluid ($CO_2$ or propane), the subcritical conditions lead to a pinch point inside the exchanger, generally at the dew point of the fluid. Therefore, for these two condensers, it is necessary to implement the following method.

**Method for the calculation of the air temperature profile and the search of the pinch point:**

The approach is based on a discretization of the condenser or gas-cooler into cells and on the assumption of an insulated heat exchanger. This allows to perform at each node i an energy balance between the hot and cold fluids (respectively the main fluid *mf* and the air):

$$\dot{M}_{air} \cdot (h_{air\ i} - h_{air\ in}) = \dot{M}_{mf} \cdot (h_{mf\ i} - h_{mf\ out})$$

$$\Rightarrow c_{p,air}(T_{air\ i} - T_{amb}) = \mu(h_{mf\ i} - h_{mf\ out}) \quad \text{with} \quad \mu = \frac{\dot{M}_{mf}}{\dot{M}_{air}}$$

$$\Rightarrow T_{air\ i} = T_{amb} + \frac{\mu}{c_{p,air}}(h_{mf\ i} - h_{mf\ out})$$

Knowing the inlet conditions of both fluids and the conditions of the main fluid along the exchanger, these of the air can be calculated iteratively, provided that the ratio $\mu$ is known. Althought in our case the mass flow rates of $CO_2$ and propane vary with the operating conditions, which will only be fully determined once the optimizer has run, the realistic assumption has been made that the ratio $\mu$ remains constant. This amout to saying that the mass flow rate of the ambient air cooling the main fluid ($CO_2$ or propane) can always be adapted to ensure a given $\mu$. A relevant value is set for this parameter, based on typical expected values of $\dot{M}_{mf}$, $\Delta h_{mf}$, $\Delta T_{air}$ and on the caracteristics of heat exchangers fitting this use found on the market.

By setting the $\mu$ value at 1/50 in the $CO_2$ gas-cooler/condenser and 1/30 in the propane condenser, it is then possible to calculate the pinch point in each heat exchanger for any operating condition. It occurs at the node $i_0$ where the temperature approach between the two fluids is the lowest. The pinch values must not be lower than the required minimum value $\Delta T_{pinch\ air}$.

**Figure 2:** Diagram of the method implemented

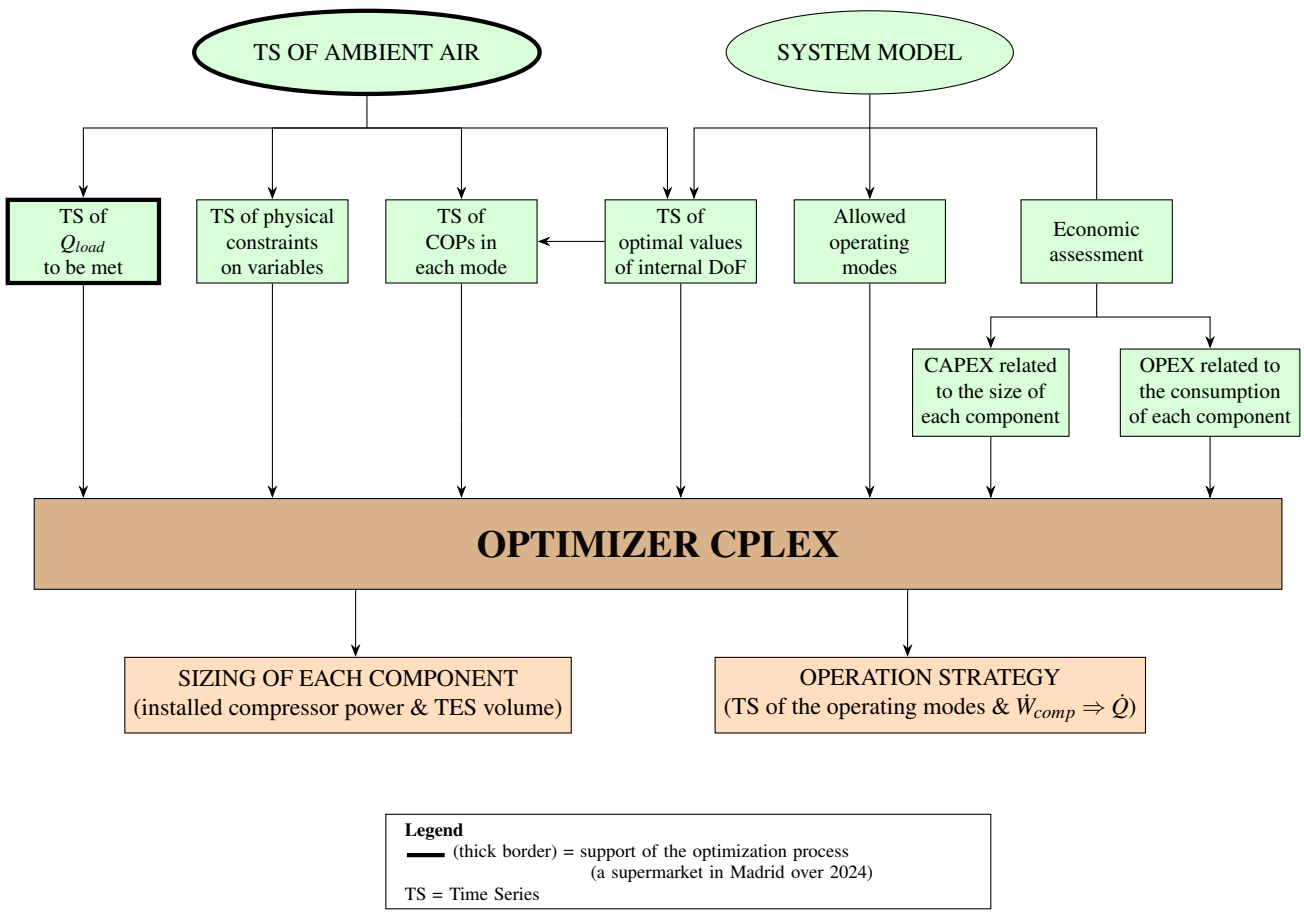


Table 3: Description of the 5 possible operating modes of the system

| MODE | DMS | TES | $\dot{Q}_{CO_2}$ | $\dot{Q}_{DMS-CO_2}$ | $\dot{Q}_{TES-CO_2}$ | $\dot{Q}_{DMS-TES}$ |
|---|---|---|---|---|---|---|
| 1.A | OFF | At rest | ✓ | × | × | × |
| 1.B | | Subcooling | ✓ | × | ✓ | × |
| 1.C | | Charging | ✓ | × | × | ✓ |
| 2.A | ON | At rest | ✓ | ✓ | × | × |
| 2.B | | Subcooling | ✓ | ✓ | ✓ | × |

### 3.2. Nature of the main cycle identification

The first step in defining the optimal parameters of the main $CO_2$ cycle is to determine whether it will operate under subcritical or transcritical conditions. This decision affects the way in which certain parameters are subsequently fixed or optimized.

It has been confirmed by modeling, that there is an ambiant air temperature limit below which operation will be subcritical and beyond which operation will be transcritical. With respect to $\Delta T_{pinch\ air}$ in the condenser/gas cooler, a dichotomous search has been implemented to determine this limit $T_{amb}^{LIM}$.

### 3.3. Optimal condenser/gas-cooler outler - $T_{gc\ out}$

In the interest of energy efficiency, the $CO_2$ cooling in contact with ambient air is always maximized, with operational limits imposed by $\Delta T_{pinch\ air}$. Thus, the optimization of the temperature $T_{gc\ out}$ consists in setting it to its lowest possible value.

- For a subcritical cycle, this value is determined by the condensation pressure, given the two phase equilibrium. Yet, the latter pressure is itself obtained with a minimization under the constraint imposed by $\Delta T_{pinch\ air}$, using a dichotomous search algorithm.

- For a supercritical cycle, it has been found that the pinch point is located at the heat exchanger outlet (where the $CO_2$ is the colder). To determine this, numerous temperature profiles in the gas cooler under appropriate operating conditions have been plotted and analyzed (using the same calculation method as developed in Section 3.1). It ensures that, regardless of the heat rejection pressure, setting $T_{gc\ out} = T_{amb} + \Delta T_{pinch\ air}$ is sufficient to achieve the minimum pinch requirement.

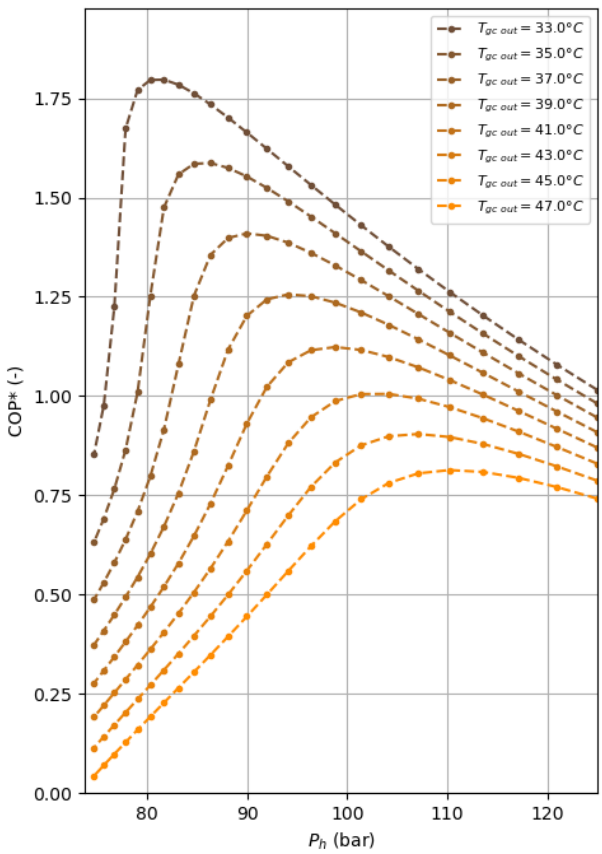
### 3.4. Optimal heat rejection - $P_h$

In accordance with some results from the literature, the models show an optimal value of the heat rejection pressure in transcritical $CO_2$ cycles. It depends on the degree of DMS.

If the $CO_2$ is not subcooled it has been found, in accordance with [15], that each value of $T_{gc\ out}$ corresponds to a certain optimal value of $P_h$ that maximizes COP* (in this unsubcooled case, COP* = $COP_{CO_2}$). Figure 3 illustrates this result. The relationship determined between these two quantities is purely linear and given below ($T_{gc\ out}$ is expressed in $°C$). The coefficient of determination of the interpolation is $R^2 = 0.9989$. Furthermore, at 38°C, the determined optimal value of $P_h$ differs by 1.82% from the one got by [15] in the same conditions.

$$P_h = 2.1821 \cdot T_{gc\ out} + 19.9625 \quad \text{[bar]}$$

Figure 3: COP* vs $P_h$ for different $T_{gc\ out}$



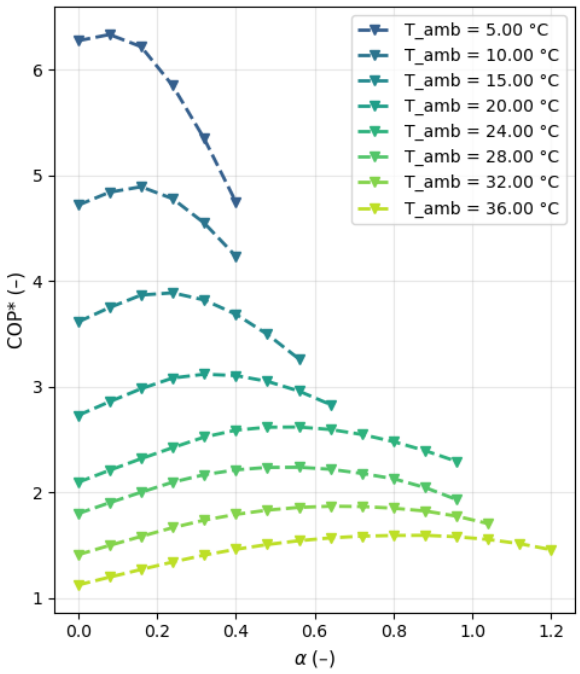

If a degree of DMS $\Delta T_{sc\ DMS}$ is added to the transcritical cycle, the optimal value of $P_h$ still exists but varies with this new variable, as mentionned by [7], [8] and [17]. Thus for each couple ($T_{gc\ out}$, $\Delta T_{sc\ DMS}$), COP* is maximized with a certain $P_h$. Using $\alpha$ as the coupling parameter, Figure 4 shows the maximal COP* obtained with this optimized value of $P_h$, for different $T_{gc\ out}$ and $\alpha$.

### 3.5. Optimal share of DMS - $\alpha$

Once $P_h$ set to its optimal or two-phase equilibrium value for each ($T_{gc\ out}$, $\alpha$) couple, it is possible to analyze the influence of the DMS share on the instantaneous overall performance COP*.

In accordance with [8] as well as [18], Figure 4 shows that at fixed $T_{amb}$ which imposes $T_{gc\ out}$, it is possible to find an optimal ratio $\alpha$ maximizing COP*.

Figure 4: COP* vs $T_{amb}$ and $\alpha$




Since there is an optimal value for $\alpha$ at each ambient temperature, it is no longer a variable on which the solver can act, but rather an input parameter. Actually, at each time step, the solver can switch to a mode without or with DMS (mode 1 or 2), but if DMS is enabled, $Q_{DMS-CO_2}$ is imposed by:

$$\dot{Q}_{DMS-CO_2} = \alpha_{opt} \cdot \dot{Q}_{CO_2} = \alpha_{opt} \cdot \Delta h_{CO_2} \cdot \dot{M}_{CO_2}$$

where $\alpha_{opt}$ and $\Delta h_{CO_2}$ result from by $T_{amb}$ and $\dot{M}_{CO_2}$ remains a real degree of freedom for the solver.

### 3.6. Maximum share of TES subcooling - $\beta$

Unlike DMS, the subcooling provided by the TES has no immediate energy cost for the system. Thus, $\beta$ is one of the variables on which the solver can actually act.

However, there is a physical limit to the value of $\beta$, imposed by the water storage temperature, hence the upper bound $\beta_{max}$:

$$\begin{cases} \beta_{max} = 0 \text{ if } T_{TES\ in} \geq T_{\text{TES hot}} + \Delta T_{\text{pinch water}} \\ \beta_{max} = \frac{h_{TES\ in} - h(P_h\ ;\ T_{\text{TES cold}} + \Delta T_{\text{pinch water}})}{\Delta h_{CO_2}} \text{ otherwise} \end{cases}$$

where $T_{\text{TES cold}} + \Delta T_{\text{pinch water}}$ is actually the colder temperature that the $CO_2$ can achieve when it is subcooled by the water.

## 4. Optimization

### 4.1. Clustering of time series

With a relevent time step of $\Delta t = 1$ h and a time horizon of one year (assuming yearly periodicity to get simulations over 10 years), the time series counts 8784 coordinates. Yet, the preprocessing required to determine the preliminary optimal parameters (see Section 4.2) involves non-linear subproblems. Thus, dimension reduction is necessary to enable calculations.

A clustering analysis was performed to identify typical days for which the input parameters would be calculated. In practice, the k-means clustering method was implemented using `cluster.KMeans` from the library `sklearn`. Given that the goal is to define an optimal system operating strategy for each typical day, it is important that the hourly temperatures for any given day of the year be as close as possible to those of the reference day for its category. As a result, the temperature time series of each day type is the group average and the standard deviation between individuals within a group is the relevant indicator for quantifying the quality of a given clustering.

Figure 5 summarizes the clustering statistics. Once all the days of the year 2024 are divided into the 3 temperature categories (*low*, *medium*, or *high*), it clearly appears that the breakdown almost corresponds to a temporal breakdown.

Additionally, the critical case (*worst day*) is defined as the day with the highest average temperature, namely July 24$^{th}$ for 2024. In these conditions, the refrigeration system has the most degraded performance which enable to size the installation.

Once the average hourly temperature profiles for each category of days are calculated (see Figure 5), they can be used as input data for the subsequent calculations.

It starts with the refrigeration load profile, another external constraints. The model specially developed in [14] for the supermarket of the case study was has been used to compute this load that the system must meet over time.

Then the electricity price signal is the third external time series to be computed. Thanks to OMIE, which manages the entirely the day-ahead electricity market for the Iberian Peninsula, it is possible to collect the day-ahead market hourly prices in Spain for all the days of the year 2024. In order to estimate the real cost (on the bill) of the electricity in €/kWh for the considered supermarket, different contributions have to be added to these wholesale prices, as detailed in Table 4 which excludes the unnecessary fixed cost of the electricity contract.

Table 4: Contributions added to the wholesale electricity prices

| CONTRIBUTION | AMOUNT | SOURCE |
|---|---|---|
| Supplier margins | 0,0165 €/MWh | average market price for SME from CNMC [19] |
| Special electricity tax | 5,11269632 % | Spanish government [20] |
| VAT | 21 % | Radenergy [21] |

### 4.2. Thermodynamic time series

With the same methods implemented to get the results presented in Section 3, the time series of the optimal thermodynamic parameters, the physical boundaries and the COPs are computed. The starting point of the calculation is $T_{amb}$, from which result the optimal values of $T_{gc\ out}$, then $P_h$ and possibly $\alpha$ (in the mode 2). Each operating mode (see Table 3) has its own thermodynamic time series. Then are enventually used as input data for the optimization problem. Table 5 summarizes

the different time series (required inputs for the solver or necessary to compute the thermodynamic cycle) calculated for each mode, while the implemented calculation algorithm is detailed in the algorithm in Section Appendix A.

| MODE | Required inputs | Intermediaries |
|---|---|---|
| 1.A | $COP_{CO_2,1}$ | $T_{amb}$ \| $T_{gc,out}$ \| $P_h$ |
| 1.B | $COP_{CO_2,1}$ \| $\beta_{max,1}$ | |
| 1.C | $COP_{CO_2,1}$ \| $COP_{DMS-TES}$ | |
| 2.A | $COP_{CO_2,2}$ \| $COP_{DMS-CO_2}$ | $T_{amb}$ \| $T_{gc,out}$ \| $P_h$ \| $\alpha$ |
| 2.B | $COP_{CO_2,2}$ \| $COP_{DMS-CO_2}$ \| $\beta_{max,2}$ | |

**Table 5:** Thermodynamic inputs by operating mode

It is worth noting that altough the ratio $COP_{CO_2}$ (defined in Section 2.3) is not changed by $\beta$, it is influenced by $P_h$, whose optimal value varies with the one of $\alpha$. Thus, one time series of $COP_{CO_2}$ is computed for each family of operating modes (modes 1 and 2).

The same difference applies to $\beta_{max}$ (defined in Section 3.6), which is calculated using:

$$\begin{cases} T_{\text{TES in}} = T_{\text{gc out}} & \text{in MODE 1 } (\alpha = 0) \\ T_{\text{TES in}} = T_{\text{DMS out}} & \text{in MODE 2 } (\alpha > 0) \end{cases}$$

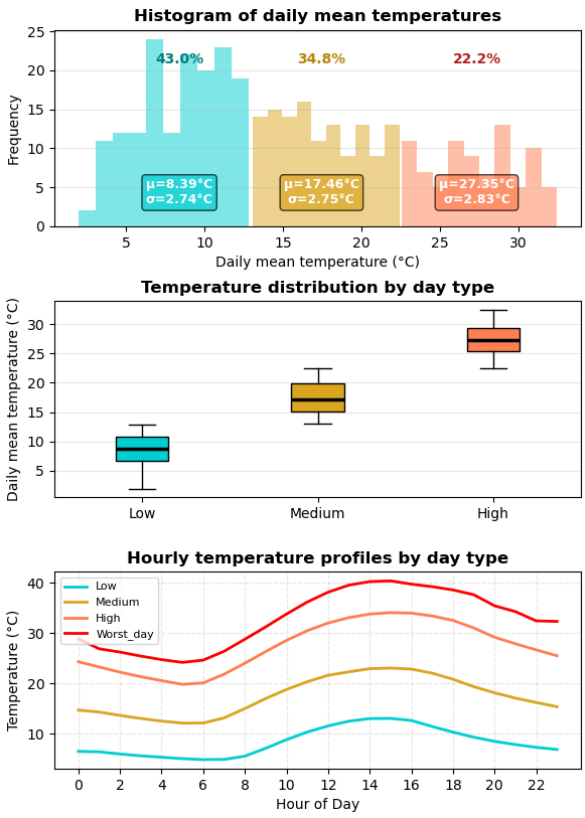



**Figure 5:** Clustering analysis suummary

### 4.3. Optimization problem formulation

The optimization problem consists in finding the optimal design and operation over time of the refrigeration system given certain external constraints (refrigeration load profile, ambient temperature, electricity prices), through a total cost minimization. The objective function is written as:

$$\begin{aligned} & f\left(\dot{W}_{\text{comp CO}_2}, \dot{W}_{\text{comp DMS}}, z_{\text{comp CO}_2}, z_{\text{comp DMS}}, v_{\text{TES}}\right) \\ & = f_{\text{CAPEX}}\left(z_{\text{comp CO}_2},\ z_{\text{comp DMS}},\ v_{\text{TES}}\right) \\ & \quad + f_{\text{NPV OPEX}}\left(\dot{W}_{\text{comp CO}_2},\ \dot{W}_{\text{comp DMS}}\right) \end{aligned}$$

with

$$\begin{cases} f_{\text{CAPEX}} = c_{CO_2}\left(z_{\text{comp CO}_2}\right) + c_{\text{DMS}}\left(z_{\text{comp DMS}}\right) + c_{\text{TES}}\left(v_{\text{TES}}\right) \\ f_{\text{NPV OPEX}} = C^T_{\text{elec}} \Delta t \left(\dot{W}_{\text{comp CO}_2} + \dot{W}_{\text{comp DMS}}\right) \sum_j \frac{1}{(1+r)^i} \end{cases}$$

The decision variables are the size of the components (the installed capacity of compressors $z_{comp}$ and the volume of storage tanks $v_{TES}$), their operating modes (binary variables) and their power levels over time ($\dot{W}_{comp}$). The constraints (listed below) include the need to constantly meet the refrigeration load, the physical limits of the system, the daily cycle of the TES, the exclusivity of the propane cycle (which cannot simultaneously cool both $CO_2$ and water) and of the TES (which cannot simultaneously be charged and discharged) or the operation under optimal thermodynamic settings (like $\alpha$, see Section 4.2).

On the one hand, the OPEX part of the cost is calculated using the time series of the electricity price signal $C_{elec}$. An assessment of the TES pump power consumption has shown it was 40 to 400 times lower than the consumption the of the compressors. The estimation is detailed in **Supplementary material - TES OPEX Assessment**. This ontribution will therefore be neglected. On the other hand, the CAPEX part calculation requires a size-related cost of the components ($c_{CO_2}$, $c_{DMS}$ and $c_{TES}$), which are assumed to be affine. The relations used and their determination methods are summarized in Table 6 and detailed in **Supplementary material - CAPEX Assessment**. In contrast to the CAPEX, which is assumed to be a one-time initial investment, the OPEX is discounted over the lifetime of the system, set to X years, with a discount rate $r = 0.08$.

Table 6: Summary of CAPEX evaluation methods for the different system components

| Component price evaluation function | Price evaluation method |
|---|---|
| $C_{CO_2} \approx 11151.4 \cdot \dot{W}_{comp} + 10103$ [€] | The total investment cost for the $CO_2$ cycle includes compressor, gas cooler and evaporators. Compressor and gas-cooler cost are estimated using correlations from [22]. Since they have been established for heat pump cycles, the results are adapted to refrigeration applications using a corrective multiplicative factor equal to 1.5. The evaporators are actually refrigerated display cases, whose cost are provided by a supplier. Thermal sizing assumptions (COP, logarithmic mean temperature differences (LMTD) and heat transfer coefficients) are used to express the component costs as a function of compressor power. The resulting nonlinear cost function is linearized over the operating range (50–175 kW). |
| $C_{DMS} \approx 1873.6 \dot{W}_{comp} + 14413$ [€] | The total investment cost for the propane cycle includes compressor, condenser and evaporator. Power-law correlations reported in [17] are used. Heat exchanger areas are estimated from simplified thermal sizing (based on assumptions about COP, heat transfer coefficients and LMTD). The total CAPEX is expressed as a function of compressor electrical power and then linearized over the operating range (4–12 kW). |
| $C_{TES} \approx 800 \cdot v_{TES}$ [€] | The specific investment cost of this linear relation derives from reported European projects that are compiled by the Celsiuscity-eu initiative in the 2015 Thermal Energy Storage report. The cost per $m^3$ of water equivalent is adjusted to account for a two-tank configuration and additional heat exchanger (for discharging when subcooling the $CO_2$). |

**Constraints of the optimization problem:**

$$\forall i, \quad \dot{W}^i_{\text{comp DMS}} \leq z_{\text{comp DMS}}$$
$$\forall i, \quad \dot{W}^i_{\text{comp CO}_2} \leq z_{\text{comp CO}_2}$$
$$\forall i, \quad V^i_{\text{TES hot}} \leq v_{\text{TES}}$$
$$\forall i, \quad V^i_{\text{TES cold}} \leq v_{\text{TES}}$$
$$|V^0_{\text{TES hot}} - V^N_{\text{TES hot}}| \leq \varepsilon$$
$$|V^0_{\text{TES cold}} - V^N_{\text{TES cold}}| \leq \varepsilon$$
$$\dot{Q}_{load} = \dot{Q}_{\text{DMS}-\text{CO}_2} + \dot{Q}_{\text{TES}-\text{CO}_2} + \dot{Q}_{CO_2}$$
$$\dot{Q}_{\text{TES}-\text{CO}_2} = c_{p,w} \cdot \Delta T_w \cdot \dot{M}_{\text{w disch}}$$
$$\dot{Q}_{\text{DMS}-\text{TES}} = c_{p,w} \cdot \Delta T_w \cdot \dot{M}_{\text{w ch}}$$
$$\forall i, \quad 0 = \dot{Q}^i_{\text{TES}-\text{CO}_2} \cdot \dot{Q}^i_{\text{DMS}-\text{TES}}$$
$$\forall i, \quad 0 = \dot{Q}^i_{\text{DMS}-\text{CO}_2} \cdot \dot{Q}^i_{\text{DMS}-\text{TES}}$$
$$\forall i, \quad \dot{M}^i_{\text{w,c}} = \max\left(\frac{\rho_w}{\Delta t}(V^{i+1}_{\text{TES hot}} - V^i_{\text{TES hot}}),\ 0\right) = \max\left(\frac{\rho_w}{\Delta t}(V^i_{\text{TES cold}} - V^{i+1}_{\text{TES cold}}),\ 0\right)$$
$$\forall i, \quad \dot{M}^i_{\text{w,d}} = \max\left(\frac{\rho_w}{\Delta t}(V^i_{\text{TES hot}} - V^{i+1}_{\text{TES hot}}),\ 0\right) = \max\left(\frac{\rho_w}{\Delta t}(V^{i+1}_{\text{TES cold}} - V^i_{\text{TES cold}}),\ 0\right)$$

**In MODE 1** where $\dot{Q}^i_{\text{DMS}-\text{CO}_2} = 0$ :

$$\dot{W}^i_{\text{comp,CO}_2} = \frac{1}{\text{COP}^i_{\text{CO}_2,1}} \dot{Q}^i_{\text{CO}_2}$$
$$\dot{W}^i_{\text{comp,DMS}} = \frac{\dot{Q}^i_{\text{DMS}-\text{CO}_2} + \dot{Q}^i_{\text{DMS}-\text{TES}}}{\text{COP}^i_{\text{DMS}-\text{CO}_2}}$$
$$\dot{Q}^i_{\text{TES}-\text{CO}_2} \leq \beta^i_{\text{max},1} \dot{Q}^i_{\text{CO}_2}$$

**In MODE 2** where $\dot{Q}^i_{\text{DMS}-\text{CO}_2} > 0$ :

$$\dot{W}^i_{\text{comp,CO}_2} = \frac{1}{\text{COP}^i_{\text{CO}_2,2}} \dot{Q}^i_{\text{CO}_2}$$
$$\dot{W}^i_{\text{comp,DMS}} = \frac{\dot{Q}^i_{\text{DMS}-\text{CO}_2} + \dot{Q}^i_{\text{DMS}-\text{TES}}}{\text{COP}^i_{\text{DMS}-\text{CO}_2}}$$
$$\dot{Q}^i_{\text{TES}-\text{CO}_2} \leq \beta^i_{\text{max},2} \dot{Q}^i_{\text{CO}_2}$$
$$\dot{Q}^i_{\text{DMS}-\text{CO}_2} = \alpha^i \dot{Q}^i_{\text{CO}_2}$$

### 4.4. *Linearization of the constraints*

The computation of numerous optimal parameters through non-linear optimization prior to the final optimization problem eases its linearization of the problem. Yet, in order to run a linear solver, some constraints still need to be reformulated.

Mixed-Integer Linear Programming (MILP) is used, particularly for handling the mode selection with the binary variables $b_{\text{MODE}}$. For instance, the implementation of the exclusivity between the two subcooling sources gives:

$$\dot{Q}^i_{\text{TES}-\text{CO}_2} \cdot \dot{Q}^i_{\text{DMS}-\text{TES}} = 0$$
$$\Rightarrow \begin{cases} \dot{Q}^i_{\text{TES}-\text{CO}_2} \leq M(1 - b_{\text{MODE}}) \\ \dot{Q}^i_{\text{DMS}-\text{TES}} \leq M \cdot b_{\text{MODE}} \end{cases}$$

Here $M$ is a sufficiently large constant enabling to implement the Big M method which is commonly used to extend the simplex algorithm. Thus, the following calculation of $\dot{M}_{\text{w ch}}$, the mass flow rate of water during the charge of the TES, activated by the binary variable $b_{\text{ch}}$, shows how maximum functions can be reformulated:

$$\begin{aligned}\dot{M}^{i}_{\text{w ch}} &= \max\left(\frac{\rho_w}{\Delta t}(V^{i+1}_{\text{TES hot}} - V^{i}_{\text{TES hot}})\ ,\ 0\right)\\ &= \max\left(\frac{\rho_w}{\Delta t}(V^{i}_{\text{TES cold}} - V^{i+1}_{\text{TES cold}})\ ,\ 0\right)\\ \Rightarrow &\begin{cases}\dot{M}^{i}_{\text{w ch}} \geq 0\\ \dot{M}^{i}_{\text{w ch}} \geq \frac{\rho_w}{\Delta t}(V^{i+1}_{\text{TES hot}} - V^{i}_{\text{TES hot}})\\ \dot{M}^{i}_{\text{w ch}} \leq 0 + M\cdot b_{\text{ch}}\\ \dot{M}^{i}_{\text{w ch}} \leq \frac{\rho_w}{\Delta t}(V^{i+1}_{\text{TES hot}} - V^{i}_{\text{TES hot}}) + M(1-b_{\text{ch}})\\ \dot{M}^{i}_{\text{w ch}} \geq \frac{\rho_w}{\Delta t}(V^{i}_{\text{TES cold}} - V^{i+1}_{\text{TES cold}})\\ \dot{M}^{i}_{\text{w ch}} \leq \frac{\rho_w}{\Delta t}(V^{i}_{\text{TES cold}} - V^{i+1}_{\text{TES cold}}) + M(1-b_{\text{ch}})\end{cases}\end{aligned}$$

For this latter set of constraints involving two successive indices, the boundary condition results from the cyclicity of the TES operation, requiring that the volume of water in the reservoirs be sufficiently close at the beginning and end of each day.

## 5. Results

Once the linear problem is formulated, it is possible to solve it with the optimizer `ILOG CPLEX` from `IBM`. Several configurations have been compared: the reference system without any subcooling device, the system with DMS only and the system combining DMS and TES.

### 5.1. First optimization results

The first comparison of the three configurations highlights the predominance of the DMS system. First, implementing DMS alone halves the installed $CO_2$ compressor capacity. This latter is the predominant initial expenditure, so the CAPEX is strongly reduced (-46%) and the annual energy consumption is lowered by 15%. Second, adding TES to DMS provides an additional, though smaller, improvement. Compared to the reference system, the DMS-only configuration decreases the total cost by 43.5% and the energy consumption by 15.4%, while the combined configuration achieves nearly the same energy savings with a 46.8% cost reduction. The marginal gain resulting from the combination is obtained with a limited storage volume (2.8 $m^3$). Actually, the dominant benefit comes from subcooling via DMS, with TES only used to shave the remaining peaks in refrigeration demand.

### 5.2. Capping the DMS share $\alpha$

The method implemented pre-calculates the optimal $\alpha$ for each time step, based on an optimization of the instantaneous overall system performance (COP*). However, since TES enables to shift the energy consumption of the system over time, it is interesting to cap the pre-computed DMS share $\alpha$ to explore the performance of the system when TES is more significantly mobilized. Therefore, the time series of the optimal DMS share $\alpha$ (see Section 3.5) are recalculated with a maximum value set to $\alpha_{\text{max}}$. Additionally to the previously obtained very small volume of TES, this approach is motivated by the major capacity of the propane cycle got (29% of the installed $CO_2$ cycle capacity). In fact, regulations limit the maximum charge of propane in commercial refrigeration systems [23].

The outcomes of this sensibility study are coherent. On the one hand, the installed capacity of the expensive $CO_2$ compressor $z_{\text{comp CO}_2}$ remains nearly constant (78 kW). On the other hand, the TES volume falls as DMS share increases (from nearly 80 m³ at 0.20 to less than 3 m³ in the uncapped configuration). This suggests that higher DMS participation allowed substitutes for storage capacity rather than complementing it. The techno-economic balance is reached when DMS and TES share the improvements achieved.

The approach is fruitful since the total cost almost plateaus beyond a certain level of capping. The Figure 6 shows that for $\alpha_{\text{max}}$ (the upper bound on $\alpha$) higher than 0.45, this plateau is reached, while the optimal installed volume of TES plumets. Thus, compared to the uncapped configuration, setting $\alpha_{max} = 0.45$ enables a paramount increase in $V_{\text{TES}}$ (multiplied by 13.8) and a significant decrease in $z_{\text{comp DMS}}$ (divided by 2.6), without any variation in total cost. Moreover, this capping value is the one that allows the system to consume the minimum amount of energy. Indeed, consumption is on the rise for both lower and higher values of $\alpha_{max}$.

### 5.3. Sensitivity to TES temperature strategy

Beyond $\alpha_{max}$, the water storage temperature in another hyperparameter of the model that can influence the optimal design and operation found. Actually, given a constant temperature difference between the two tanks ($\Delta T_{\text{TES}} = 5\,\text{K}$), the storage temperatures $T_{\text{TES cold}}$ directly determines the maximum subcooling that the TES can provide to the $CO_2$ through the upper bound $\beta_{\text{max}}$ (see in Section 3.6), and so the electricity consumption that can be shifted over time.

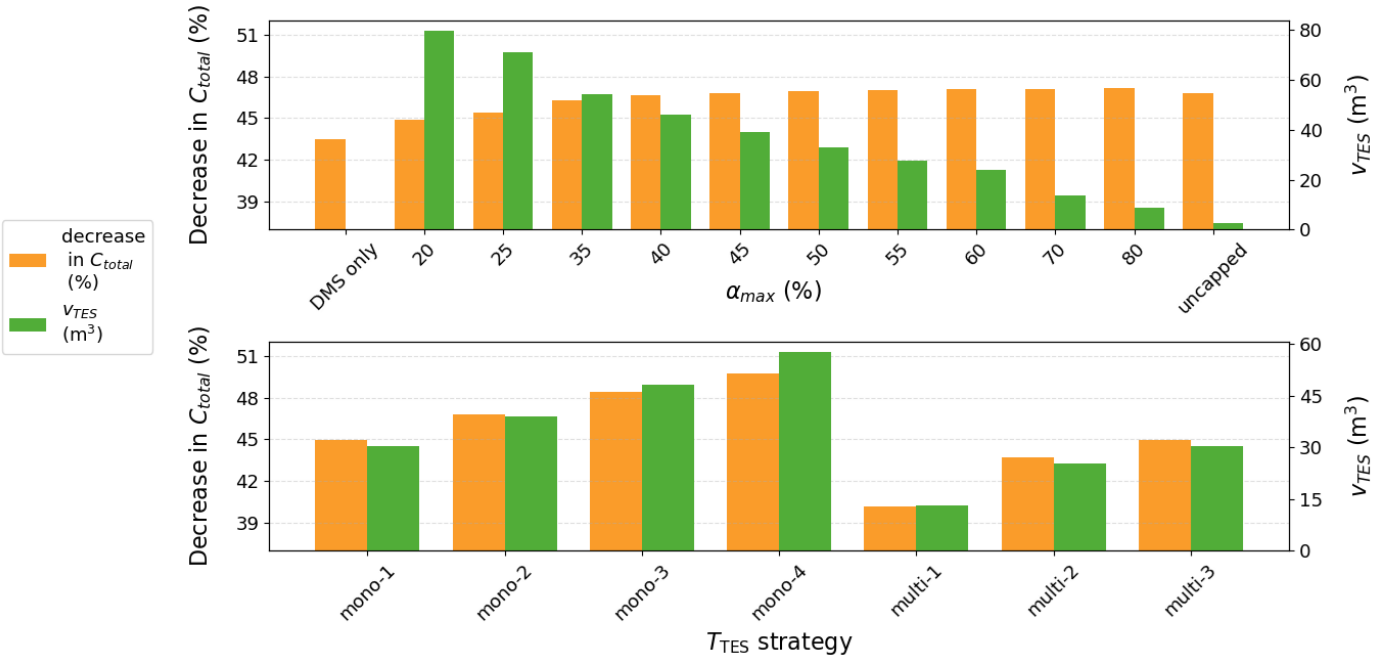
Thus, a sensitivity analysis is conducted. The different cases compared are detailed in Table 7. Two main strategies are assessed, each one with several temperature levels:

- **Mono-temperature strategy:** Same storage temperatures $T_{\text{TES}}$ maintained for all days, whatever $T_{\text{amb}}$.
- **Multi-temperature strategy:** Storage temperatures $T_{\text{TES}}$ adapted to the ambient temperature (through the day types).

The motivation behind the multi-temperature strategy is that with a lower gap between $T_{\text{amb}}$ and $T_{\text{TES cold}}$, the efficiency of the propane cycle during the charging phase of the TES is improved. Yet the results show that any of these adaptive temperature strategy incurs a cost penalty relative to the mono-temperature options (see Figure 6), with the most extended adaptation (multi-1) being the most unfavorable.

Indeed, the gain in charging performance is not sufficient to compensate for the loss in discharging capacity. This latter loss results from the reduction in maximum subcooling that the TES can provide on the warmer days (with higher $T_{\text{TES cold}}$) when it is most needed (high ambient temperature and electricity prices). The same reason explains why lowering the TES

**Figure 6:** Sensitivity of $C_{total}$ and $v_{TES}$ to the capping of the DMS share $\alpha_{max}$ (top) and the TES temperature strategy (bottom)




temperature constantly decreases the total cost of the system. The colder the storage temperature, the higher $v_{TES}$, which enables to shift more energy consumption and so to reduce the total cost. Storing thermal energy at the lowest possible temperature maximizes the available subcooling ($\beta_{max}$) during the critical discharge phases. Thus, the mono-temperature configurations show a reduction of the total cost by 8.7% when $T_{TES\ cold}$ is decreased from 7°C to -8°C, and the multi-temperature configurations share this trend.

A more in-depth analysis of the optimal hourly operating pattern in each compared case shows that the decrease in TES temperature enables to systematically combine subcooling from both the TES and the DMS system. At each time step, mode 2.B prevails if the TES is not being charged. In contrast, with the higher storage temperatures, mode 1.B (with subcooling from the TES only) and mode 2.A (with DMS only) are used alternatively for the days when the ambiant temperature is not high enough to combine the two subcooling sources. In this case, the pre-computed optimal value of $\alpha$ implies that the $CO_2$ temperature outing the possible DMS would be too cold to be further subcooled by the TES. As a result, the solver chooses between using the DMS or partly discharging the TES.

The total cost varies by less than 4% between the different values of $\alpha_{max}$ explored within a given TES temperature strategy, whereas the same cost varies by almost 20% between the configurations used to implement the TES. Therefore, the performance are far more sensitive to the TES temperature strategy than to the the capping of the DMS share $\alpha$. The sensitivity analysis confirms that the first one governs the baseline cost level, while the DMS maximum allowed share distributes the installed capacity of the two subcooling systems to reach this cost level.

**Table 7:** TES temperature strategies: mono vs. multi configurations

| Day type | $T_{TES}$ (°C) | mono | | | | multi | | |
|---|---|---|---|---|---|---|---|---|
| | | **1** | **2** | **3** | **4** | **1** | **2** | **3** |
| Low | $T_{TES\ cold}$ | 7 | 2 | -3 | -8 | 2 | 2 | 2 |
| | $T_{TES\ hot}$ | 12 | 7 | 2 | -3 | 7 | 7 | 7 |
| Medium | $T_{TES\ cold}$ | 7 | 2 | -3 | -8 | 7 | 4 | 2 |
| | $T_{TES\ hot}$ | 12 | 7 | 2 | -3 | 12 | 9 | 7 |
| High | $T_{TES\ cold}$ | 7 | 2 | -3 | -8 | 12 | 7 | 7 |
| | $T_{TES\ hot}$ | 12 | 7 | 2 | -3 | 17 | 12 | 12 |
| Worst day | $T_{TES\ cold}$ | 7 | 2 | -3 | -8 | 17 | 10 | 7 |
| | $T_{TES\ hot}$ | 12 | 7 | 2 | -3 | 22 | 15 | 12 |

### 5.4. *Ultimate optimal strategy, design and operation*

According to the previous results, a focus is made on the strategy mono-4 with a DMS share capped at $\alpha_{max} = 0.45$. The negative TES temperature can be achieved using a glycol water mixture to prevent freezing.

The optimal design of the system and the resulting performance over 10 years are given in Table 8. Then, the optimal operation of the system is illustrated in Figure 7 with the time series of the load distribution among the $CO_2$ cycle and the different subcooling sources (DMS and TES).

Table 8: Comparison of optimal design and performance between configurations

| Indicator | Reference | Configurations | | *VS reference* |
|---|---|---|---|---|
| | | DMS only | DMS + TES | |
| $z_{\text{comp CO}_2}$ (kW) | 166.0 | 83.9 | 71.5 | −49% / −57% |
| $z_{\text{comp DMS}}$ (kW) | 0 | 24.5 | 9.2 | — / — |
| $v_{\text{TES}}$ (m$^3$) | 0 | 0 | 65.5 | — / — |
| $C_{\text{capex}}$ (k€) | 1861.7 | 1005.5 | 891.3 | −46% / −52% |
| $C_{\text{npv opex}}$ (k€) | 162.2 | 137.8 | 132.6 | −15% / −18% |
| $C_{\text{total}}$ (k€) | 2023.9 | 1143.3 | 1023.9 | −44% / −49% |
| $w_{\text{tot}}$ (MWh) | 245.1 | 207.2 | 200.0 | −15% / −18% |

## 6. Limitations and perspectives

The system modeling developed in this study has its limitations, which mostly result from the simplifications adopted to make optimization computable.

The most significant simplifications are made in the implementation of TES, with neglected heat loss or even a two-tank solution instead in the single tank exhibiting a temperature gradient. Since the storage temperature influences the actual installation, it would also be relevant to take this temperature into account in the cost evaluation function of this component $c_{TES}$ which currently only depends on the volume $v_{TES}$.

Analysis of the solutions shows that, as soon as it is physically possible, the optimizer prioritizes the combined use of the two subcooling devices. It would therefore be interesting to assess the impact of letting the solver choose the order of the two subcooling devices. Although it would degrade the COP of the propane cycle, since it would cool an already partially subcooled $CO_2$, placing the latter at the outlet of the exchanger with the water would allow the storage temperature of the TES to be raised. Yet if the two heat exchangers are swapped, the temperature of the $CO_2$ at the inlet of the DMS would depend on the undetermined subcooling provided by the TES ($\beta$). Consequently, the value of $\alpha$ could no longer be pre-calculated as it is currently done. As a result, such a modification requires a nonlinear optimizer for the ultimate phase.

Moreover, the currently calculated optimal values of the DMS share $\alpha$ maximize the instantaneous overall performance (COP*), which does not take into account the benefits of the time lag in energy consumption enabled by TES. This is highlighted by the results provided by the capping $\alpha_{max}$. Consequently, using nonlinear optimization would also be relevant to make $\alpha$ a true degree of freedom for the solver. Energy performance would then be fully optimized on a daily timescale.

Finally, it is important to emphasize the strong sensitivity of this study to CAPEX evaluation functions. The lack of manufacturer data increases the degree of uncertainty associated with these functions ($c_{CO_2}$, $c_{DMS}$ and $c_{TES}$).

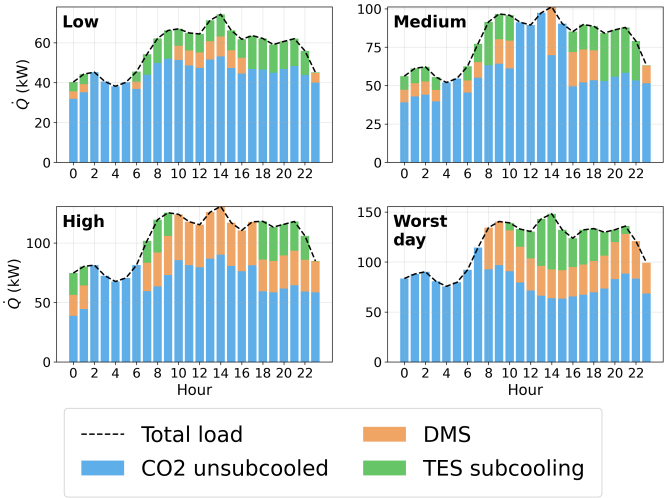



Figure 7: Hourly load distribution - Optimal configuration

## 7. Conclusions

This modeling and optimization work led to the development of a comprehensive method for measuring the potential for performance improvement in a $CO_2$ refrigeration system using subcooling devices. This method combines detailed thermodynamic modeling, cost assessment, nonlinear pre-calculation and linear optimization to determine both the optimal sizing and the optimal control strategy over a representative year of the complex system. This latter combines a small mechanical vapour-compression cycle as well as a thermal energy storage system that are both coupled to the main $CO_2$ cycle.

Applied to a supermarket refrigeration case in Madrid, the results unambiguously demonstrate the predominance of DMS in improving the energy and economic performance of the system. Implementing DMS alone achieves a 15.4% reduction in annual energy consumption and a 43.5% reduction in total cost over 10 years.

The combination of DMS and TES provides further benefits, achieving a 18.4% energy reduction and a 49.4% total cost reduction compared to the reference configuration. The annual performance coefficient reached 3.20, representing a 19.1% increase compared to the baseline configuration.

The storage temperature strategy strongly influences performance of the system, with lower TES temperatures consistently yielding better results. However, the marginal gains from TES integration are modest when DMS operates according to the share $\alpha$ maximizing the overall instantaneous performance. The sensitivity analysis reveals that TES plays a more significant role when this share is capped. Specifically, setting $\alpha_{\text{max}} = 0.45$ enables a nearly 14-fold increase in TES volume while almost halving the DMS compressor capacity, without affecting the total cost. This configuration also achieves the minimum energy consumption among all studied scenarios.

Thus, thanks to TES, the system can be designed with a limited DMS installation, without significantly compromising the economic performance. Not only can limiting the size of the DMS cycle be aligned with regulations, but it also allows

for further significant reductions in the consumption peak of the system. From 168 kW for the reference configuration, it achieves 109 kW (-35 %) if DMS only is implemented and 78 kW (-54 %) with the combined configuration. This result is particularly interesting in the context of grid strain at certain critical times during the hottest days.

## Appendix A. Computation algorithm for the thermodynamic time series

The Figure A.8 presents the algorithm implemented to compute the time series of the thermodynamic parameters that are part of the optimizer inputs. They are the physical boundaries ($\beta_{max}$) and the performances (COPs) of each cycle. The starting point of the calculation is $T_{amb}$, from which results the optimal values of $T_{gc\ out}$, then $P_h$ and possibly $\alpha$ (in the MODE 2). Then the thermodynamic cycles can be computed and the required parameters are deduced. Two different time series are calculated for the two families of operating modes (see Table 3): one for the modes without DMS (1.A, 1.B and 1.C) and one for the modes with DMS (2.A and 2.B).

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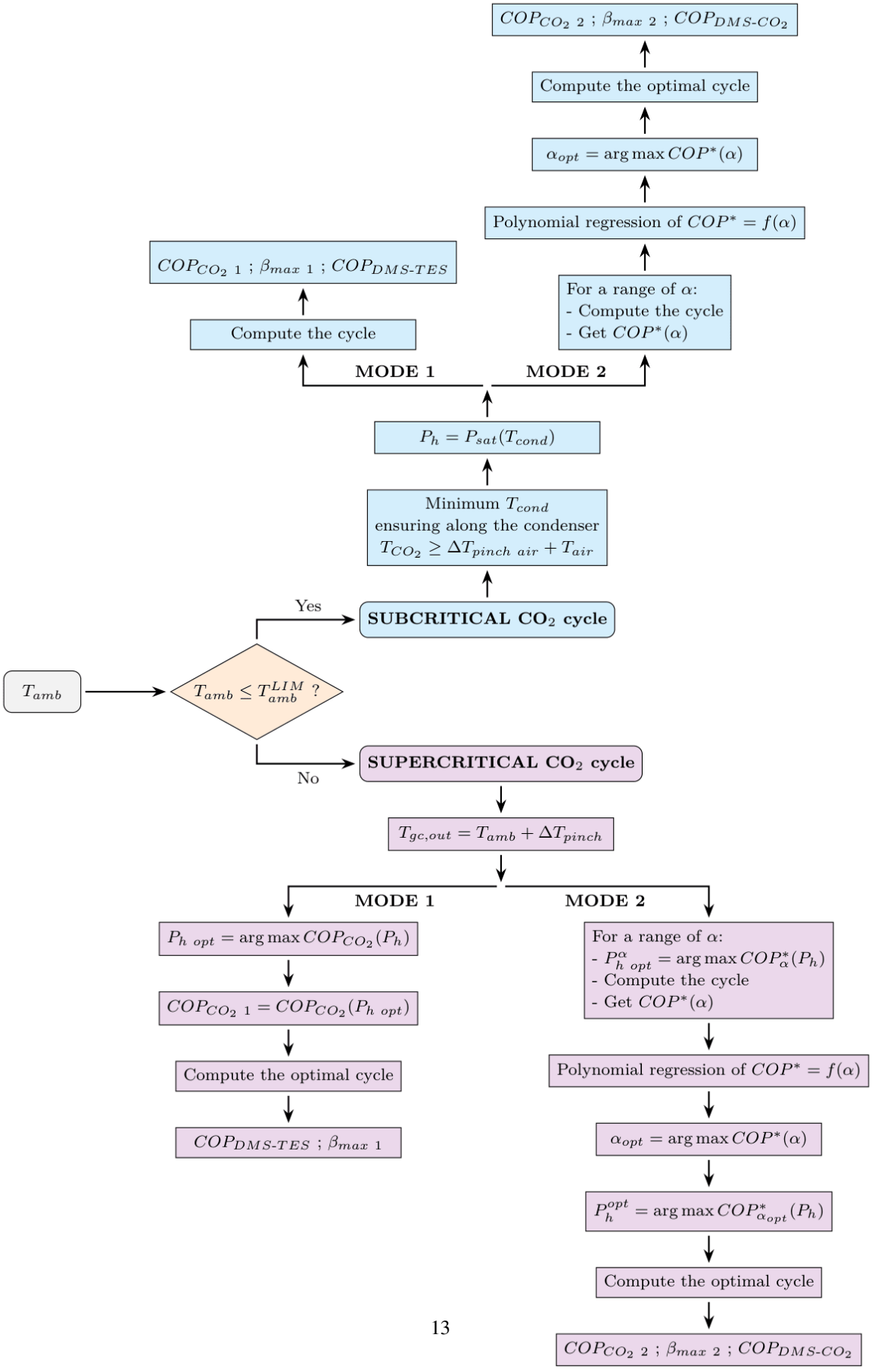
**Figure A.8:** Algorithm implemented to compute the thermodynamic time series



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